Print Hollow Ractangle Pattern

```
* * * * *
*       *
*       *
* * * * *
```

	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	
(1,1)	*	*	*	*	*	1
(2,1)	*				*	2
(3,1)	*				*	3
(4,1)	*	*	*	*	*	4
	1	2	3	4	5	

[4 x 5]

① Total lines (Total rows)
↳ outer (1 to 4)

② (row = 1 || col = 1 || row = 4 || ~~row~~ col = 5)
↳ Inner loop

code:-

```
public class patterns {
    public static void hollow_rec(int totalRows, int totalcols){

        //outer loop
        for(int i=1; i<= totalRows ; i++){
            // inner loop
            for( int j=1 ; j<= totalcols ; j++){
                //cell - (i,j)
                if (i==1|| i==totalRows ||j==1|j==totalcols)
                {        //boundry cell
                    Sop ("*");
                } else{
                    Sop(" ");
                }
            } Sopln ();
        } }
    }
```

```
public static void main (String [] a){
        ~~hollow~~ hollow_rec (4, 5);
    }
}
```

→ O/P :-

```
* * * * *
*       *
*       *
* * * * *
```

☐ Dry Run :-

i=1	j = ~~1 2 3 4 5~~ 6	Output
i=2	j = ~~1 2 3 4 5~~ 6	* * * * *
		* _ _ _ *
i=3	j = ~~1 2 3 4 5~~ 6	* _ _ _ *
		* * * * *
i=4	j = ~~1 2 3 4 5~~ 6	
i=5	×	

☐ Inverted & Rotated Half PYRAMID :-

*				*	r=1	Space=3	Star=1
* *			*	*	r=2	2	2
* * *		*	*	*	r=3	1	3
* * * *	*	*	*	*	r=4	0	4

① lines (outer loop) → 4
② row no => same star
③ Space => 4 - i (row)

code

```
public class Patterns{

    public static void invertedRHP (int n){

        for(int i=1; i<=n ; i++){

            for(int j=1 ; j<= n-i ; j++){
                Sop (" ");
            }
            for(int j=1 ; j<= i ; j++){
                Sop("*");
            }
            Sop ln();
        }
    }
    public static void main (String [] a){

        invertedRHP (4);
    }
}
```

☐ Inverted Half - PYRAMID with Numbers :-

1 2 3 4 5	$i=1$	1 to 5	$\rightarrow n-i+1$
1 2 3 4	$i=2$	1 to 4	$\rightarrow n-i+1$
1 2 3	$i=3$	1 to 3	$\rightarrow n-i+1$
1 2	$i=4$	1 to 2	
1	$i=5$	1 to 1	

① Lines - outer loop : 5 times
② inner loop : Start 1 go to $n-i+1$

Code :-

```
public class Pattern {
    public static void inverted_Half_withNumbers (int n) {

        for(int i=1; i<=n; i++) {

            for(int j=1; j<= n-i+1; j++){

                Sop ( j+ "  ");
            }

            Sopln ();
        }
    }

    public static void main (String [] a){

            inverted_Half_withNumber (5);
    }
}
```

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FLOYD's Tringle

1					$i=1$
2	3				$i=2$
4	5	6			$i=3$
7	8	9	10		$i=4$
11	12	13	14	15	$i=5$

① outer loop => n = 5 (lines) int n = 5;

② Inner loop int counter = 1;

↳ How many time counter prints

Code:-

```
public class {
    public static void floyds_traingle (int n) {
            int counter = 1;
        for(int i=1; i<=n; i++){  //outer
            //inner - for how many times will counter be print
            for(int j=1; j<=i ; j++){

                Sop(counter+" ");
                Counter ++;
            }
            Sopln();
        }
    }
    public static void main (String [] a){
            floyds_traingle (5);
    }
}
```

0-1 Triangle :-

	(1,1) 1					$i=1$	$j=1$ to 1
(2,1)	0	(2,2) 1				$i=2$	$j=1$ to 2
(3,1)	1	(3,2) 0	(3,3) 1			$i=3$	$j=1$ to 3
(4,1)	0	(4,2) 1	(4,3) 0	(4,4) 1		$i=4$	$j=1$ to 4
(5,1)	1	(5,2) 0	(5,3) 1	(5,4) 0	(5,5) 1	$i=5$	$j=1$ to 5

① Outer loop : n times
② Printed in each line
↳ for(int j=1; j<=i; j++)
③ (row + col)
(i+j) → even = "1"
(i+j) → odd = "0"

Code

```
public class Solution{
    public static void zero_one(int n){
        for (int i=1; i<=n; i++){
            for(int j=1; j<=i; j++){

                if((i+j) % 2 == 0){

                    Sop("1");
                }
                else{
                    Sop("0");
                }
            }
        }
    }
    public static void main(String [] a){
        zero_one(5);
    }
```

Butterfly Pattern :-

$n = 4$

```
*       *      i=1  star=1   Spaces = 6 = 2x3 (n-i)
* *     * *    i=2  star=2   Sp = 4     = 2x2 (n-i)
* * *   * * *  i=3  star=3   Sp = 2     = 2x1 (n-i)
* * * * * * *  i=4  star 4   Sp = 0     = 2x0 (n-i)
* * * * * * *
* * *   * * *
* *     * *
*       *
```

1st half :- ① outer loop

for(i=1 ; i<=n ; i++)

② line

star + spaces + stars

$i + 2\times(n-i) + i$

2nd half :- ① Outer loop

for(i=n ; i>=1 ; i--)

② Inner loop

↳ same as first half

Code :-

```
public class Pattern {

    public static void butterfly (int n) {
```

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```
//1st half
for (int i=1; i<=n; i++){

    // star - i
    for(int j=1; j<=i ; j++){

            Sop ("*");
    }

    // Space - 2*(n-i)
    for( int j=1; j<= 2*(n-i) ; j++){

            Sop (" ");
    }

    // star - i
    for(int j=1 ; j<= i ; j++){

            Sop("*");
    }

    Sopln();
}

//2nd half
for (int i=n; i>=1 ; i--){

    // star - i
    for(int j=1 ; j<=i ; j++) {
            Sop ("*");
    }
```

```
        // Spaces - 2*(n-i)
        for(int j=1; j<= 2*(n-i); j++){

                Sop(" ");
        }

        // Star - i
        for(int j=1; j <= i; j++){

                Sop("*");
        }
        Sopln();
    }
}

public static void main(String [] a){

        butterfly(4);
    }
}
```

- Solid Rhombus:- $n=5$

				*	*	*	*	*	$i=1$	Spaces $=4(n-i)$	Star $=5$
			*	*	*	*	*		$i=2$	$sp=3(n-i)$	$st=5$
		*	*	*	*	*			$i=3$	$sp=2(n-i)$	$st=5$
	*	*	*	*	*				$i=4$	$sp=1(n-i)$	$st=5$
*	*	*	*	*					$i=5$	$sp=0(n-i)$	$st=5$

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① outer loop : for(int i=1; i<=5; i++)

② inner loop : space + star = line (i)

Code :-

```
public class Patterns{
    public static void solid_rhombus (int n){
        for (int i=1; i<=n; i++){
            for (int j=1; j<=(n-i); j++){
                Sop(" ");
            }
            for (int j=1; j<=n; j++){
                Sop("*");
            }
            Sopln();
        }
    }
    public static void main (String [] a){
        Solid_rhombus (5);
    }
}
```

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Hollow Rhombus :-

			*	*	*	*
		*			*	
	*			*		
*	*	*	*			

① Outer loop
↳ for (int i = 1 to 4)

② Line
Spaces + boundary rectangle
(n-i) + hollow rectangle

Code :-

```
public class Pattern {
    public static void hollow (int n) {
        //outer
        for( int i=1; i<=n; i++) {
            // Spaces
            for (int j=1; j<=(n-i); j++) {

                Sop (" ");
            }

            //hollow rectangle - Stars.

            for ( int j=1 ; j<=n; j++) {
```

```
            if(i==1 || i==n || j==1 || j==n){
                    Sop("*");
            }
            else{
                    Sop(" ");
            }
        }
        Sopln();
    }
}

public Static void main (String [] a){
        hollow (4);
    }
}
```

- Diamond Pattern :-

$n = 4$

*	$i=1$	$sp = 3\ (n-i)$	$st = 1$	$(2i-1)$
* * *	$i=2$	$sp=2$	$st=3$	$(2i-1)$
* * * * *	$i=3$	$sp=1$	$st=5$	$(2i-1)$
* * * * * * *	$i=4$	$sp=0$	$st=7$	$(2i-1)$
* * * * * * *				
* * * * *				
* * *				
*				

1st half :-

① outer loop
↳ for (int i = 1; i<=n; i++)

② what to print
↳ (2i-1) :- Space + Star
(n-i) + (2i-1)

2nd half :-
i = n to 1
& i--

Code :-

```
public class Pattern {

    public static void diamond (int n) {

        //1st half
        for (int i=1; i<=n ; i++) {

            //Spaces
            for(int j=1; j<= (n-i) ; j++) {

                Sop(" ");
            }

            //Star
            for(int j=1; j<= (2*i)-1; j++) {

                Sop("*");
            }
```

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```
        Sopln();
    }

    //2nd half:-

    for(int i=n; i>=1; i--){

        //Spaces
        for(int j=1; j<=(n-i); j++){

            Sop(" ");
        }

        //Stars
        for(int j=1; j<=(2*i)-1; j++){

            Sop("*");
        }

        Sopln();
    }
}
public static void main(String [] a){

        diamond (4);
    }
}
```

⇒ Bonus live

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Number PYRAMID

		numbers	
_ _ _ _ 1	$i=1$	1 1x	$sp=4 \rightarrow (n-i)$
_ _ _ 2 2	$i=2$	2 2x	$sp=3$
_ _ 3 3 3	$i=3$	3 3x	$sp=2$
_ 4 4 4 4	$i=4$	4 4x	$sp=1$
5 5 5 5 5	$i=5$	5 5x	$sp=0$

① outer loop : 1 to n(5)

② what to print

Space + numbers

```
for (j=1 to i)
    { Print (i + " ");
```

```
for (int j=1 to (n-i)) { sop (" ");}
```

Code

```
public class Pattern {

    public static void numberPyramid (int n) {

        //outer loop:
        for(int i=1 ; i<=n ; i++){
            //spaces
            for (int j=1 ; j<=n-i ; j++) {
```

```
            Sop("  ");
        }
        //numbers
        for(int j=1;j<=i;j++){
            Sop(j+"  ");
        }

        Sopln();
    }
}
public static void main(String [] a){
    numberPyramid(5);
}
}
```

- PALINDROMIC Pattern with numbers:-

1	$i=1$	$sp=4$
2 1 2	$i=2$	$sp=3$
3 2 1 2 3	$i=3$	$sp=2$
4 3 2 1 2 3 4	$i=4$	$sp=1$
5 4 3 2 1 2 3 4 5	$i=5$	$sp=0$

① Outer loop → (1 to n)

② what to print :

Space + Number

↓ ↓ ↘

$sp=(n-i)$ decending acending

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	decending	ascending
i = 1	1 to 1	X
i = 2	2 to 1	2 to 2
i = 3	3 to 1	2 to 3
i = 4	4 to 1	2 to 4
i = 5	5 to 1	2 to 5
	↓	↓
	i to 1	2 to i
	↓	↓
	for(int j=1; j >=1; j--)	for(int j=2; to i)

code:-

```
public class Patterns {

    public static void Palindrom (int n) {

        for( int i=1; i<=n; i++) {

            for(int j= 1; j<=n-i ; j++){

                sop(" ");
            }

            for (int j= i; j >=1 ; j--){

                Sop (j);
            }
```

```
        for(int j=2 ; j<=i ; j++){
            Sop(j);
        }

        Sopln();
    }
}
public static void main (String [] a){
        Palindrom.(5);
    }
}
```